

MATTER-FIRST DISCOVERY OF SELF-CONSISTENT WARP-PRECURSOR SPACETIMES WITH GRTECLYN

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We present an automated, *matter-first* search for warp-precursor spacetimes that are genuine solutions to the Einstein equations. MAP-Elites (quality-diversity) and CMA-ES propose sub-luminal scalar shell configurations; **GRTresna** solves the constraint equations; **GRTEclyn** evolves the spacetimes in full 3+1 numerical relativity on GPUs; and a *4D evolving null-geodesic tracer* falsifies candidates at high resolution. Our primary result, candidate **eval 144**, is a dynamic five-lump exotic shell whose end-to-end 4D shortcut is $f_{\text{geo}} = 7.96\%$ (5/5 rays, Hamiltonian quality OK) at HQ validation ($N = 256$, $t = 30$), matching the search-tier measurement to $\sim 0.1\%$. Frozen mid-run geodesics peak higher (11.5%) but collapse by $t = 30$; the 4D trace is the authoritative transport certificate. No matter moves superluminally; this is a transient, exotic-matter-supported warp *precursor*, not a transport drive.

I. INTRODUCTION

Warp spacetimes are usually studied *metric-first*: one prescribes $g_{\mu\nu}$, reads off $T_{\mu\nu} = G_{\mu\nu}/8\pi$, and analyzes frozen slices [?]. Such matter is rarely obtained from a consistent initial-value problem and need not survive evolution. Here “FTL” means an *effective* null shortcut relative to flat space between the same asymptotic endpoints—never local superluminal motion outside the light cone.

We invert this: the optimizer proposes matter; **GRTresna** returns compatible initial data; **GRTEclyn** evolves it; probes ask whether a gauge-invariant null shortcut emerges and survives falsification.

A. Methodological Rigor

Three pillars distinguish this work from metric-first studies:

1. **Initial value problem:** constraint-clean data from physical scalar sources, not coordinate scripts.
2. **Dynamic feedback:** full BSSN/CCZ4 evolution; dispersing matter collapses the channel.
3. **4D geodesic falsification:** the headline metric is **ftl_geo_evolving**—null rays integrated through the *time-evolving* metric stack. Frozen-slice geodesics can over- or under-state transport (positive control **eval 144** vs. negative control **eval 086**).

B. Contributions

- (i) End-to-end pipeline with matched QD/CMA-ES objective and HQ 4D verify (Secs. ??–??).

- (ii) Candidate **eval 144**: verified $\sim 8\%$ 4D null shortcut at HQ resolution (Sec. ??).

- (iii) Honest limits: transient precursor, exotic matter, late-time trapping—not passenger transport (Sec. ??).

II. MATTER-FIRST PIPELINE

A. Search Loop

$\theta \rightarrow$ initial data $\rightarrow$ **GRTEclyn** $\rightarrow$ diagnostics $\rightarrow S(\theta)$. Scalar sources [Eq. (??)] feed **GRTresna**; the score combines 4D shortcut terms, health (constraints, survival), and risk (exotic/energy conditions). In the **ftl_4d** campaign the headline channel is **ftl_geo_evolving** from the 4D tracer; frozen geodesic and coordinate FTL are gated off until the trace completes.

$$\rho = \frac{1}{2}\psi^{-4}|\tilde{\nabla}\phi|^2 + \frac{1}{2}\Pi^2 + V(\phi), \quad S_i = -\Pi\partial_i\phi, \quad (1)$$

B. Shell Parameterization

A low-dimensional shell vector defines scalar clouds on a Fibonacci sphere with searchable orientation, toroidal/poloidal/radial currents, exotic wedge, and mass/coupling. MAP-Elites preserves diverse mechanism families across an 8×8 archive (**ftl_lifetime** descriptor).

C. Self-Consistency

Matched signed-scalar evolution, mass-matched potential, and a post-load gate ($\|H\|_{L^2}, \|M\|_{L^2} \lesssim 10^{-2}$) reject inconsistent **.gridinit** files before GPU evolution.

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TABLE I. Search hand-off summary (ftl_4d line).

stage	eval	score	4D f_{geo}
QD best	156	508	7.0%
CMA-ES winner	144	596	8.0%
HQ falsified	144	283	7.96%

D. Three-Stage Search (ftl_4d)

- **Stage 1 — MAP-Elites QD (ftl_4d_v1):** $N = 128$, $ml = 2$, $t = 16$, ~ 192 evals, 4D *search* geodesic, *ftl_first*. Best: *eval* 156, *score* 508.
- **Stage 2 — CMA-ES (ftl_4d_cmaes_v1):** warm-start from QD top-5, $\sigma_0 = 0.05$, 144 evals, same objective. Winner: *eval* 144, *score* 596 (+17% vs. QD).
- **Stage 3 — HQ falsification:** fresh GRTresna + GPU at $N = 256$, $ml = 3$, $t = 30$, 4D *hq* trace, frames, incremental scoring.

III. SEARCH METHODOLOGY

A. MAP-Elites quality-diversity

MAP-Elites illuminates an 8×8 behavior archive: one elite per (*ftl_lifetime*, *ftl_geo_evolving*) cell. The campaign ran 192 evaluations (105 GPU-ok) before hand-off to CMA-ES. Archive gains were sparse late, but a large jump at *eval* 156 (*score* 508.5, *ftl_geo_evolving*=0.35) defined the elite basin for refinement.

B. CMA-ES local refinement

CMA-ES warm-started from the QD top-5 ($\sigma_0 = 0.05$, 18 generations, 144 evals) improved the best score monotonically ($514 \rightarrow 596$, +17%) without changing resolution or objective—validating the QD→CMA-ES handoff on the 4D-scored line.

HQ *score* 283 is lower than *search* 596 mainly due to *structural_persistence* ≈ 0.63 and instability penalties over $t = 30$, not failure of the 4D shortcut (Sec. ??).

IV. DIAGNOSTICS AND PROBE VALIDATION

A. Operational and gauge-invariant FTL

Coordinate operational shortcut F_{op} detects tilted light cones but is partly gauge dependent. The authori-

tative probe integrates null geodesics,

$$\dot{x}^\mu = g^{\mu\nu} k_\nu, \quad \dot{k}_\mu = -\frac{1}{2}(\partial_\mu g^{\alpha\beta})k_\alpha k_\beta, \quad (2)$$

on the *evolving* 3+1 metric. Define

$$F_{\text{op}} = \frac{t_B^{\text{flat}} - t_B^{\text{curved}}}{t_B^{\text{flat}}}, \quad (3)$$

$$f_{\text{geo}} = \max\left(0, \frac{t_{\text{flat}} - t_{\text{min}}}{t_{\text{flat}}}\right), \quad (4)$$

with Hamiltonian null-constraint gating ($h_{\text{rel}} < 10^{-2}$). In HQ mode, five rays are traced from $t_{\text{emit}} = 0$ through the full cached metric stack (~ 124 slices).

B. Probe controls

Prescribed Alcubierre ($f_{\text{geo}} \approx 32\%$) and Minkowski ($f_{\text{geo}} = 0$) calibrate the codebase. Candidate *eval* 086 is the negative control: frozen peak $\sim 5.8\%$ but 4D HQ trace $\rightarrow 0$ (0/5 rays, quality fail). *eval* 144 is the positive control: 4D gain survives HQ.

V. RESULTS: VERIFIED 4D WARP PRECURSOR (EVAL 144)

A. Provenance and genome

eval 144 is the CMA-ES winner (Stage 2), warm-started from QD *eval* 156. The genome is a *dynamic* five-lump exotic shell ($\sim 91\%$ exotic fraction): Fibonacci sphere layout, momentum-carrying currents, shift-driven motor (*shell_static* ≈ 0). HQ replay uses a fresh GRTresna solve at $N = 256$, three refinement levels, $t = 30$, 126 frames.

B. 4D falsification certificate

The HQ evolving-geodesic artifact reports:

4D f_{geo}	7.96%
rays reached	5/5
$t_{\text{emit}} / t_{\text{arrival}} / t_{\text{flat}}$	0.00 / 13.25 / 14.40
$h_{\text{quality_ok}}$	true ($h_{\text{rel}} \approx 8 \times 10^{-4}$)
search-tier 4D f_{geo} (CMA-ES)	7.97% (3/3 rays)

The search and HQ measurements agree to $\sim 0.1\%$ relative—the 4D objective was not a resolution artefact. Effective asymptotic signal speed $\approx c/(1 - f_{\text{geo}}) \approx 1.09c$ between fixed endpoints; locally all rays remain on the light cone.

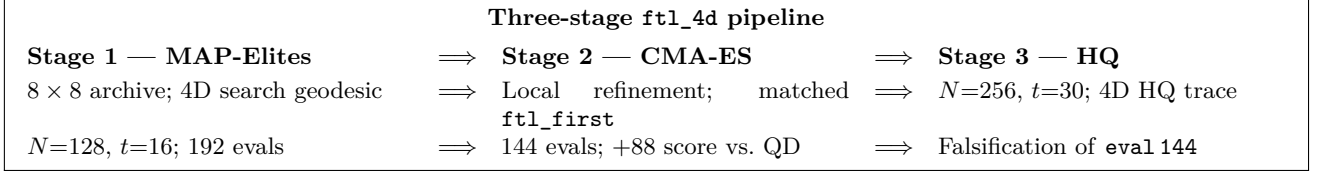


FIG. 1. Discovery pipeline for campaign `ftl_4d`. Candidate `eval 144` (Stage 2 winner) is promoted through Stage 3 (Sec. ??).

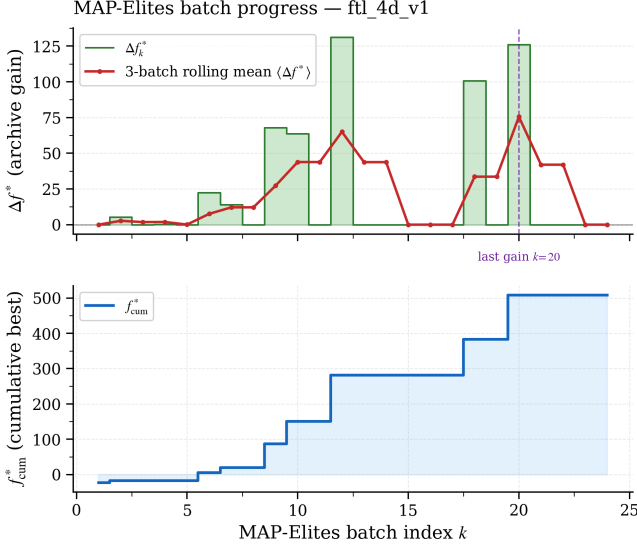


FIG. 2. MAP-Elites QD progress (`ftl_4d_v1`): batch scores and archive coverage vs. evaluation index. Saturation late; `eval 156` is the discovery peak passed to CMA-ES.

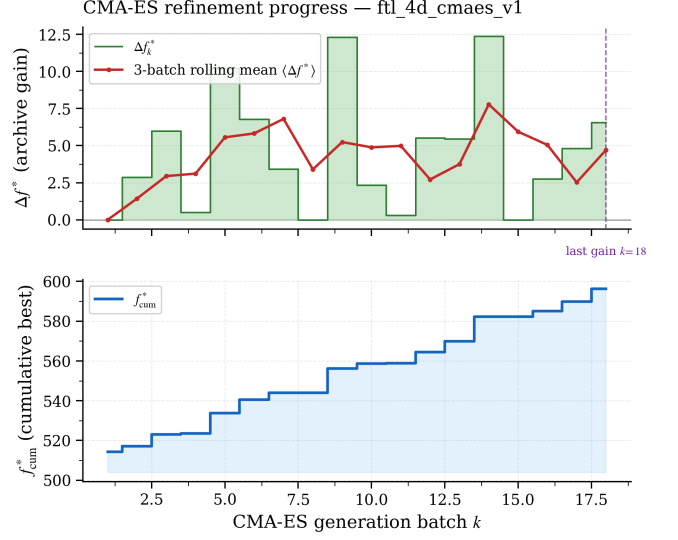


FIG. 3. CMA-ES generation progress (`ftl_4d_cmaes_v1`): best and mean batch scores vs. generation. Steady improvement; `eval 144` is the Stage 2 winner promoted to HQ.

C. Frozen vs. evolving geodesics

Table ?? contrasts *frozen-slice* diagnostics with the 4D certificate. Frozen f_{geo} peaks at 11.5% ($t \approx 5.8$) but falls to 0% by $t = 30$. Scoring frozen snapshots alone would over-credit mid-run transport and falsely declare “no FTL” at late times. The 4D trace is the only honest end-to-end witness on this dynamic bubble.

D. What moves faster than light

Nothing physical moves superluminally. Scalar clouds and local light cones remain sub-luminal; coordinate $\max c \approx 1.31$ is down-gated as delocalized coordinate offset in scoring. The $\sim 8\%$ figure is a *null-geodesic timing advantage* through the warped region relative to flat space—the same class of effect as in Alcubierre-type metrics, but here from evolved matter fields and certified by 4D integration through the full history.

VI. DISCUSSION

A. Interpretation

`eval 144` demonstrates that matter-first search with 4D falsification can discover self-consistent spacetimes whose *honest* null transport gain survives HQ promotion. Methodologically, the pairing with `eval 086` establishes when frozen geodesics fail. Physically, the candidate is a **transient warp precursor**: exotic matter (`exotic_penalty` = -1.6), dynamic bubble (`structural_persistence` ≈ 0.63), corroborated late-time trapping ($t \approx 29$) though horizon penalty suppressed—not a persistent corridor or transport vehicle.

B. Applications and next steps

The pipeline supports (i) inverse design of null-transport under energy-condition penalties, (ii) QD atlases of mechanism families, and (iii) Pareto maps of shortcut vs. exotic cost vs. persistence. Priorities: raise persistence without losing 4D f_{geo} , world-tube objectives, and non-dispersing matter models (e.g. boson stars).

TABLE II. HQ time evolution of **eval 144** ($N = 256$, $t = 30$). Frozen f_{geo} : per-frame null probe on static slices (diagnostic). 4D row: end-to-end evolving trace (authoritative).

t/M	frozen f_{geo}	max c	superlum. slice	note
0.0	0.0%	1.27	26%	no shortcut at $t = 0$
2.9	7.6%	1.28	49%	channel rising
5.8	11.5%	1.31	91%	frozen peak (overstates)
12.0	1.8%	1.17	98%	decay
16.1	0.0%	1.15	87%	frozen shortcut gone
30.0	0.0%	1.16	49%	3/5 frozen rays reach
4D end-to-end ($t_{\text{emit}} = 0$): $f_{\text{geo}} = 7.96\%$, 5/5 rays, $h_{\text{quality_ok}} = \text{true}$				

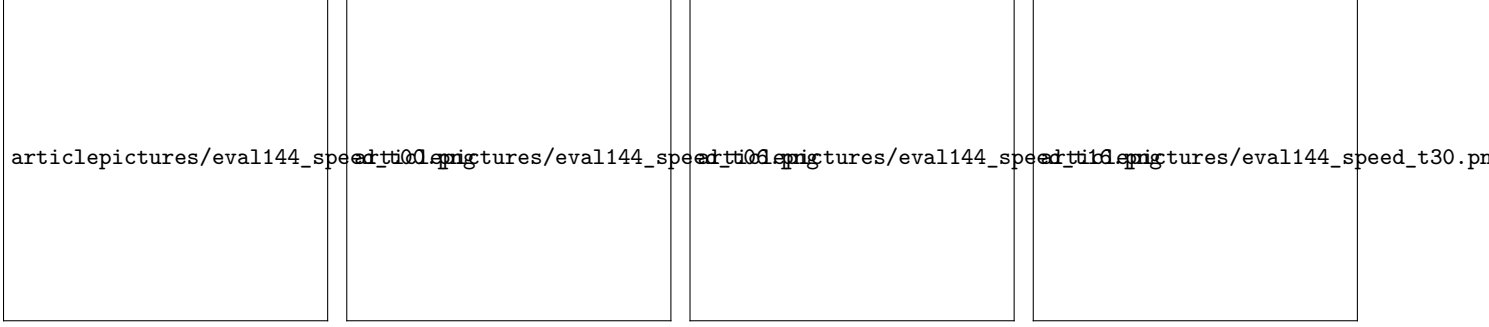


FIG. 4. Coordinate light speed on the $z = 0$ slice, HQ run **eval 144**. Left to right: $t \approx 0$, 5.8 (near frozen peak), 16, 30. Broad tilted-cone region forms and dissipates; gauge-invariant transport is certified by the 4D trace, not by any single panel.

VII. CONCLUSION

We presented a matter-first discovery pipeline with MAP-Elites, CMA-ES, and HQ 4D null-geodesic falsification. Candidate **eval 144**—a dynamic five-lump exotic shell—yields a verified end-to-end shortcut $f_{\text{geo}} =$

7.96% at HQ resolution, matching search-tier measurement. Frozen mid-run geodesics peak higher but collapse; the 4D tracer is essential. We claim a reproducible, falsifiable numerical program for warp-precursor spacetimes, not a functional warp drive.

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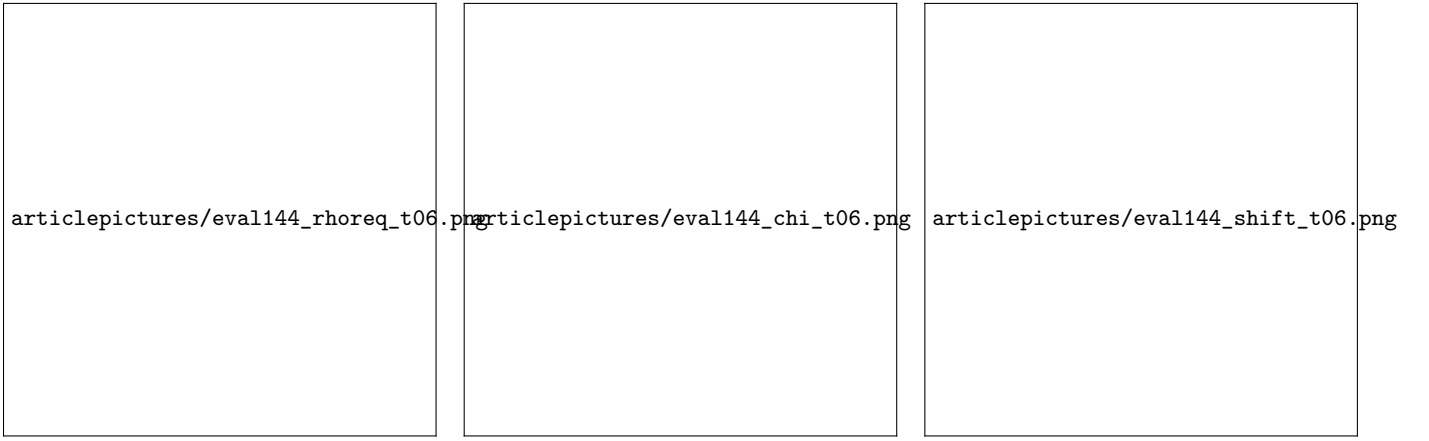


FIG. 5. Supporting fields near frozen peak ($t \approx 5.8$, $z = 0$). Left: required energy density ρ_{req} (exotic shell). Center: conformal factor χ . Right: shift component (warp motor). Dynamic, momentum-carrying source—not a static lens.